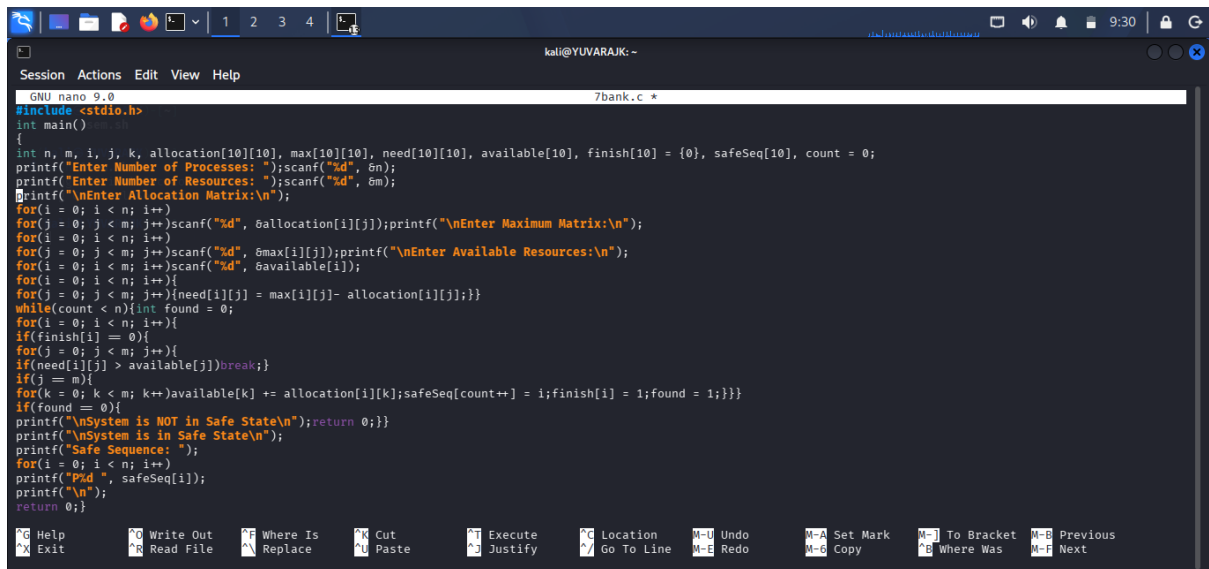
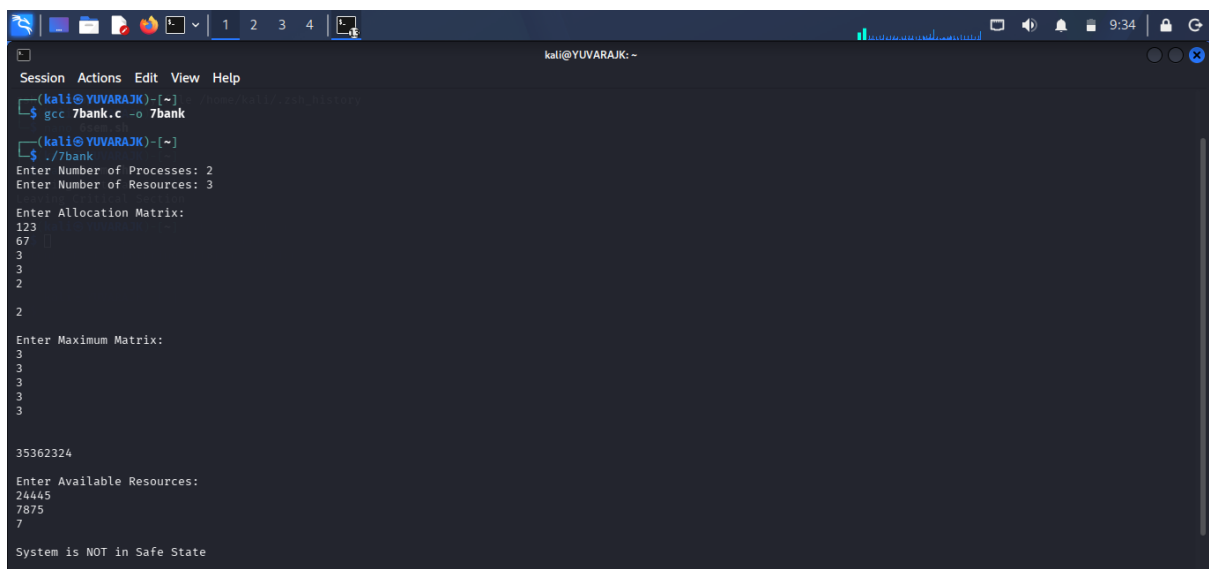


C PROGRAMMING :



```
GNU nano 9.0 7bank.c *
#include <stdio.h>
int main()
{
    int n, m, i, j, k, allocation[10][10], max[10][10], need[10][10], available[10], finish[10] = {0}, safeSeq[10], count = 0;
    printf("Enter Number of Processes: ");scanf("%d", &n);
    printf("Enter Number of Resources: ");scanf("%d", &m);
    printf("\nEnter Allocation Matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++)scanf("%d", &allocation[i][j]);printf("\nEnter Maximum Matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++)scanf("%d", &max[i][j]);printf("\nEnter Available Resources:\n");
    for(i = 0; i < m; i++)scanf("%d", &available[i]);
    for(i = 0; i < n; i++){
        for(j = 0; j < m; j++){
            need[i][j] = max[i][j] - allocation[i][j];
        }
    }
    while(count < n){int found = 0;
        for(i = 0; i < n; i++){
            if(finish[i] == 0){
                for(j = 0; j < m; j++){
                    if(need[i][j] > available[j])break;
                }
                if(j == m){
                    for(k = 0; k < m; k++)available[k] += allocation[i][k];safeSeq[count++] = i;finish[i] = 1;found = 1;
                }
            }
            if(found == 0){
                printf("\nSystem is NOT in Safe State\n");return 0;
            }
        }
        printf("\nSystem is in Safe State\n");
        printf("Safe Sequence: ");
        for(i = 0; i < n; i++)
            printf("%d ", safeSeq[i]);
        printf("\n");
        return 0;
    }
```

OUTPUT :



```
(kali@YUVARAJK)-[~]
$ gcc 7bank.c -o 7bank
(kali@YUVARAJK)-[~]
$ ./7bank
Enter Number of Processes: 2
Enter Number of Resources: 3

Enter Allocation Matrix:
123
67
3
3
2

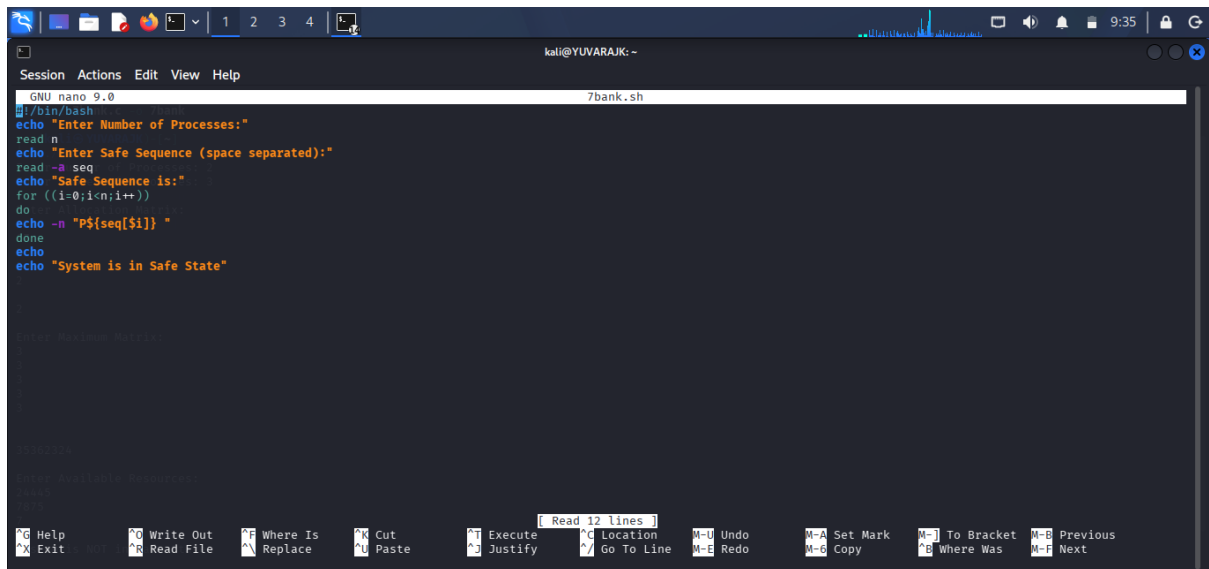
Enter Maximum Matrix:
3
3
3
3
3

35362324

Enter Available Resources:
24445
7875
7

System is NOT in Safe State
```

SHELL PROGRAMMING :

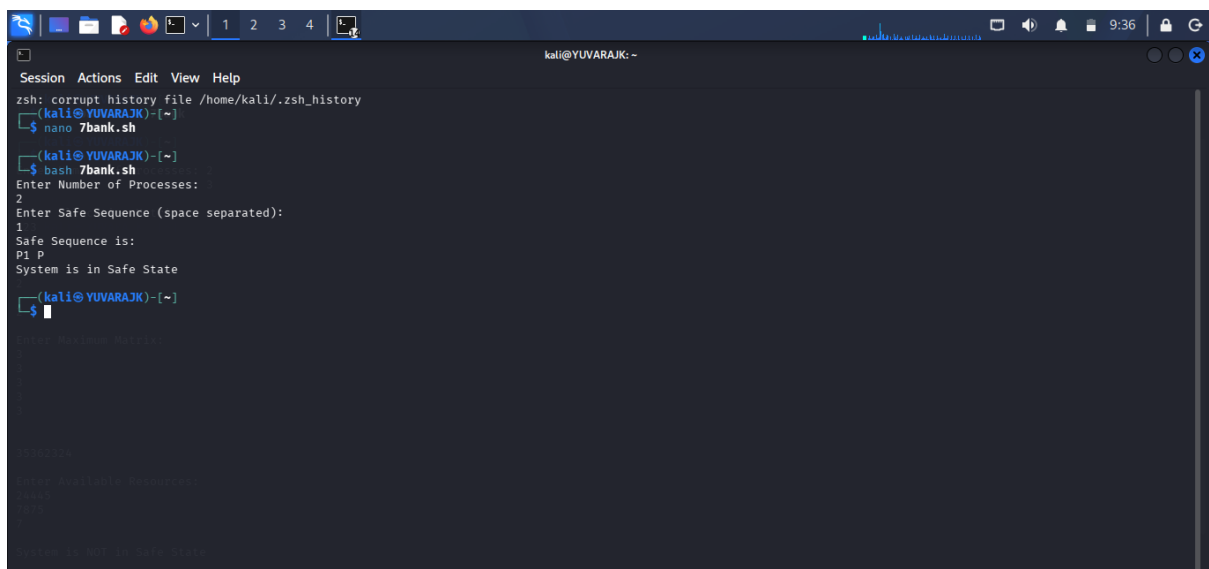


The screenshot shows a terminal window with the nano text editor open. The editor is editing a file named 7bank.sh. The script content is as follows:

```
#!/bin/bash
echo "Enter Number of Processes:"
read n
echo "Enter Safe Sequence (space separated):"
read -a seq
echo "Safe Sequence is:"
for ((i=0;i<n;i++))
do
echo -n "${seq[$i]} "
done
echo
echo "System is in Safe State"
```

The nano editor interface includes a menu bar (Session, Actions, Edit, View, Help) and a status bar at the bottom with various shortcuts like G Help, X Exit, O Write Out, R Read File, W Where Is, N Replace, C Cut, P Paste, E Execute, J Justify, L Location, G Go To Line, U Undo, R Redo, A Set Mark, C Copy,] To Bracket, B Previous, and _ Next.

OUTPUT :



The screenshot shows a terminal window where the script 7bank.sh has been executed. The output is as follows:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano 7bank.sh
(kali@YUVARAJK)-[~]
$ bash 7bank.sh
Enter Number of Processes:
2
Enter Safe Sequence (space separated):
1
Safe Sequence is:
P1 P
System is in Safe State
(kali@YUVARAJK)-[~]
$
```

The terminal window shows the user navigating to the script file, running it with 'bash 7bank.sh', and providing input for the number of processes (2) and the safe sequence (P1 P). The script then prints 'System is in Safe State'.